A coupled panel method and hydraulic network for porous airfoil aerdynamics⋆,⋆⋆

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ABSTRACT

Porous airfoils can modify aerodynamic forces through passive, pressure-driven flow between connected surface pores. This study develops a reduced-order aerodynamic–hydraulic framework for a NACA 0018 airfoil by coupling an inviscid source–vortex panel method with rectangular-channel flow models. Three configurations are examined: horizontal channels connecting chordwise locations on the same surface, vertical channels connecting the pressure and suction surfaces, and a combined arrangement of hydraulically independent channels. The predicted aerodynamic response depends strongly on channel orientation and connectivity. Horizontal channels redistribute surface pressure and increase the pressure-based lift coefficient at positive angles of attack, with larger channel heights producing stronger changes. Vertical channels promote pressure equalisation between the two surfaces and reduce the pressure-based lift relative to the solid airfoil. In the combined arrangement, the channels interact through the external pressure field, producing a response closer to the solid-airfoil baseline. Pressure and velocity difference fields indicate that flow modifications are concentrated near the surface pores. These findings highlight the importance of channel placement and connectivity in porous-airfoil design. The framework provides an efficient tool for preliminary screening, with the predicted inviscid trends requiring subsequent viscous simulation and experimental validation.

1. Introduction

Improving airfoil performance without relying on powered control systems remains an important objective in aerospace and energy applications. Conventional high-lift and flow-control devices can be effective, but they may introduce mechanical complexity, additional mass, maintenance requirements, and changes that disrupt the smooth external surface of the airfoil. Passive flow-control concepts provide an alternative strategy in which natural pressure differences around an aerodynamic body drive flow redistribution. Porous surfaces, permeable regions, localised pores, and internal channels employ this principle by permitting fluid transport between regions of unequal pressure. Their effectiveness, however, depends on pore placement, the connections between pores, and the strength of the interaction between the resulting surface-normal flow and the external aerodynamic field.

Porosity does not necessarily improve aerodynamic performance. Porosity extending along the entire airfoil chord, as well as other distributed porous arrangements, can reduce the rate of lift increase with angle of attack and produce competing effects on drag [1, 2]. Experiments and numerical studies further show that performance depends on the location of the porous region and the permitted internal flow directions [3, 4, 5]. These findings motivate the use of selectively connected pores to control pressure redistribution.

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For an airfoil containing individual channels, the aerodynamic effect depends on which pressure regions are connected. A channel connecting pores on the same airfoil surface transfers fluid between different chordwise locations, whereas a channel connecting the pressure and suction surfaces permits flow between the two sides of the airfoil. These arrangements can produce different changes in surface pressure even when their pore areas are similar. Consequently, porosity alone does not specify the design: channel connectivity and resistance must also be represented. This distinction is particularly relevant when several channels operate simultaneously, because their influence on the external pressure field can alter the flow through neighbouring channels.

Thin-airfoil models describe steady and unsteady porous-airfoil behaviour by relating pressure differences to flow through distributed permeability [6, 7], although accuracy can decrease for small porous regions or low flow resistance [8]. These formulations provide efficient aerodynamic predictions but do not explicitly resolve airfoil thickness or finite-length channels connecting selected surface pores. Their local porous boundary conditions are useful for distributed permeable surfaces, whereas modelling individual channels requires the pressure at each connected pore and a relation between pressure difference, channel geometry, and flow rate.

Passive internal connections have already been investigated for aerodynamic control. Acarer reported improved peak lift-to-drag ratio using passive slots [9], while Kestel et al. modified separated vortex flow through passive internal channels [10]. Bahadi et al. examined curved slots for improved performance at high angles of attack [11]. These studies establish the potential of pressure-driven internal

flow. The present work builds on this principle by developing a general reduced-order framework for multiple pairs of pores connected by independent channels, enabling different arrangements to be evaluated consistently.

Porous-region optimisation has also been investigated for static-stall control [12]. For individually connected pores, the design variables additionally include pore locations, channel dimensions, and the choice of same-surface or cross-surface connections. Changes in channel length and height modify hydraulic resistance, while changes in pore position modify the available pressure difference. Both effects must therefore be considered together. Exploring these combinations solely through viscous simulations or experiments is costly, motivating a computationally efficient model for preliminary assessment before detailed analysis of selected configurations.

Potential-flow panel methods offer an efficient means of calculating surface pressures on airfoils of finite thickness without discretising the surrounding fluid domain [13, 14, 15]. In a two-dimensional source–vortex formulation, the source strengths enforce the surface-normal boundary condition, while circulation and the trailing-edge Kutta condition determine the lifting solution. This structure accommodates porous flow by replacing the impermeability condition locally with a prescribed pore-normal velocity. The resulting surface velocities determine the pressure differences driving the internal channel flow. A panel formulation therefore provides a direct connection between airfoil geometry, pore flow, and aerodynamic forces at a cost suitable for repeated design evaluations.

Although the panel equations are linear for prescribed pore velocities, the complete aero–hydraulic problem is generally nonlinear. Surface pressure depends quadratically on velocity, and changing the channel flow changes both the pore velocities and the pressures driving that flow. This feedback requires a consistent coupled solution even when a linear laminar hydraulic-resistance relation is used. Treating channel flow as a fixed input would omit this interaction. The present framework instead updates the internal flow and external pressures until the coupled equations satisfy the prescribed convergence criterion.

Established viscous–inviscid methods combine an outer-flow solution with boundary-layer equations to represent additional aerodynamic effects [16]. XFOIL uses this approach with a panel-based inviscid solution and transition modelling [17]. Here, the external solver remains inviscid to isolate the aero–hydraulic interaction at low computational cost, and XFOIL provides a reference for the solid-airfoil baseline. This comparison assesses the external aerodynamic calculation; validation of the coupled porous model requires separate evidence for the internal flow and its interaction with the surrounding fluid.

The present study addresses the need for a reduced-order framework that explicitly represents finite-length channels between selected pairs of pores and couples their pressure-driven flow to the external aerodynamic solution. A two-dimensional source–vortex panel method with a Kutta condition is coupled iteratively to a hydraulic-resistance model for rectangular channels. Three NACA 0018 configurations are examined: four chordwise same-surface channels, four pressure-to-suction-surface channels, and a combined arrangement containing eight hydraulically independent channels. Channel heights of 8, 9, and 10 mm are assessed over angles of attack from $-5°$ to $15°$ at a chord Reynolds number of 5×10^5.

The combined arrangement permits the aerodynamic interaction between channel types to be assessed without introducing internal junctions. Although each channel is hydraulically independent, all channels influence a shared external pressure field. Comparing the three configurations therefore examines how their effects combine and whether connectivity changes the sensitivity to channel height and angle of attack.

The contribution is a consistent framework for comparing how channel connectivity affects internal flow, surface pressure, lift, pitching moment, and pressure-based streamwise force. The method supports preliminary design screening. Since it does not resolve boundary layers, skin friction, separation, or stall, its predictions are interpreted as inviscid trends requiring subsequent viscous simulation and experimental validation.

2. Methods

2.1. Modelling framework and assumptions

A two-dimensional coupled aero–hydraulic framework is used to analyse passive internal channels in a NACA 0018 airfoil. A *pore* denotes the surface region where a channel intersects the airfoil boundary, whereas a *channel* connects two pores through the airfoil interior. The external flow is represented by a steady source–vortex panel method, and each channel is described by a one-dimensional pressure-loss model per unit span.

The external flow is assumed incompressible, inviscid and irrotational. Boundary layers, skin friction, separation and stall are therefore not resolved. The internal channels are straight, smooth and of constant height, with sidewall effects neglected. Entrance turning, detailed pore-mouth flow and jet mixing are represented only through the stated modelling assumptions. The framework is intended for preliminary comparison of channel layouts.

The external and internal models are coupled through pore pressures and surface-normal velocities. Channel flow rates are adjusted until the two models are consistent, as summarised in Figure 1.

2.2. Airfoil and channel geometry

The NACA 0018 airfoil is generated from the standard analytical profile with a closed trailing edge [18]. A baseline of $N_0 =$ [insert panel count] cosine-spaced panels is used, with additional boundaries inserted at the pore edges. Each

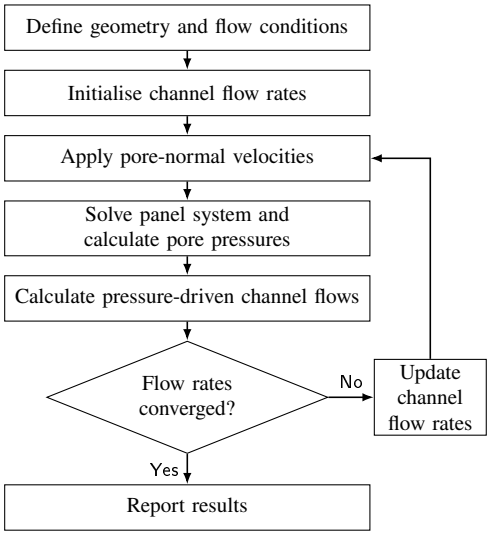


Figure 1: Coupled aero–hydraulic solution procedure. Initial flow rates are estimated from solid-airfoil pressures or a nearby converged solution. Convergence requires agreement between the current flow estimates and the pressure-driven flow rates within the prescribed tolerance.

of the resulting N panels lies entirely on a solid region or within a pore. Panel i has length S_i, midpoint (x_i, y_i), unit tangent $\boldsymbol{t}_i$ and outward unit normal $\boldsymbol{n}_i$.

Three configurations are considered:

1. **Horizontal porous airfoil:** four independent horizontal channels, with two connecting pores on the upper surface and two connecting pores on the lower surface;
2. **Vertical porous airfoil:** four independent vertical channels connecting the lower and upper surfaces at chordwise positions from approximately $x/c = 0.10$ to 0.89;
3. **Combined porous airfoil:** all eight horizontal and vertical channels, with no internal exchange of flow between them.

The layouts are shown in Figure 2. Crossings in the two-dimensional drawing do not create hydraulic connections; a physical realisation of the combined layout would require separation of the intersecting channels.

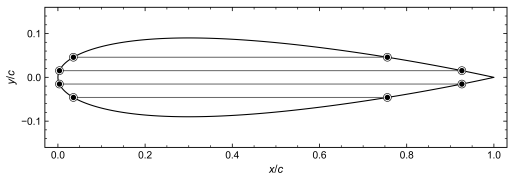
Each channel is defined by its two surface endpoints and height H_k. The centreline length L_k is the straight-line distance between these endpoints. Each pore is defined by the intersection of the channel strip with the airfoil surface. Before calculation, the geometry is checked for pore overlap, minimum pore separation and containment of the channels within the airfoil. Invalid geometries are excluded.

2.3. External aerodynamic model

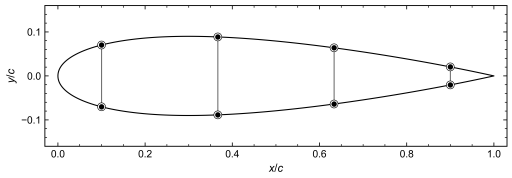
2.3.1. Source–vortex panel formulation

The panel method follows the boundary-integral representation of incompressible potential flow developed by Hess and Smith and described in later panel-method treatments [13, 19, 15]. The velocity potential Φ satisfies

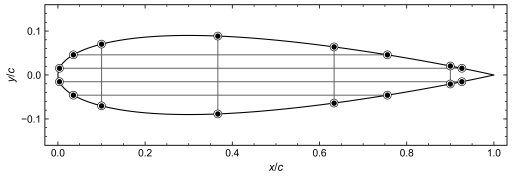
$$\boldsymbol{V} = \nabla\Phi, \qquad \nabla^2\Phi = 0, \tag{1}$$



(a) Horizontal porous airfoil



(b) Vertical porous airfoil



(c) Combined porous airfoil

Figure 2: Internal-channel layouts considered in this study.

The external velocity is represented by the freestream, one constant source strength λ_j on every panel and one uniform vortex-sheet strength γ over the closed boundary. The sources enforce the prescribed normal velocity on the finite-thickness geometry, while the vortex sheet provides circulation.

For a solid surface, the normal velocity at every control point is zero. At a pore, this boundary condition is replaced by a prescribed normal transpiration velocity, an approach consistent with reduced porous-surface formulations [6, 7]:

$$\boldsymbol{V}_i \cdot \boldsymbol{n}_i = v_{n,i}, \tag{2}$$

where $v_{n,i} > 0$ denotes flow out of the airfoil and $v_{n,i} < 0$ denotes flow into it. With the adopted influence-coefficient convention, the discrete normal equation is

$$\pi\lambda_i + \sum_{\substack{j=1 \\ j\neq i}}^{N} I_{ij}\lambda_j - \gamma \sum_{j=1}^{N} K_{ij} = -2\pi \boldsymbol{V}_\infty \cdot \boldsymbol{n}_i + 2\pi v_{n,i}, \tag{3}$$

where I_{ij} and K_{ij} are the normal source and vortex influence coefficients. The freestream vector is $\boldsymbol{V}_\infty = V_\infty(\cos\alpha, \sin\alpha)$. The first and last panels lie immediately adjacent to the trailing edge, with oppositely directed tangents. The trailing-edge Kutta condition is

$$V_{t,1} + V_{t,N} = 0. \tag{4}$$

Equations (3) and (4) form an $(N+1)\times(N+1)$ linear system for $[\lambda_1, \ldots, \lambda_N, \gamma]^{\mathrm{T}}$. Because the influence matrix depends only on the final panel geometry, it is assembled and LU-factorised once for each fixed pore-aligned discretisation. For that discretisation, changes in angle of attack or pore-normal velocity affect only the right-hand side.

2.3.2. *Surface velocity and pressure*

After solution of the panel system, the tangential velocity on panel i is

$$V_{t,i} = \boldsymbol{V}_\infty \cdot \boldsymbol{t}_i + \frac{1}{2\pi}\sum_{j=1}^{N} \lambda_j J_{ij} + \frac{\gamma}{2} - \frac{\gamma}{2\pi}\sum_{j=1}^{N} L_{ij}, \quad (5)$$

where J_{ij} and L_{ij} are the tangential source and vortex influence coefficients. The local velocity is $\boldsymbol{V}_i = V_{t,i}\boldsymbol{t}_i + v_{n,i}\boldsymbol{n}_i$. Both velocity components enter the exterior pressure closure:

$$C_{p,i} = 1 - \frac{V_{t,i}^2 + v_{n,i}^2}{V_\infty^2}, \quad (6a)$$

$$p_i = p_\infty + \frac{1}{2}\rho_\infty V_\infty^2 C_{p,i}, \quad (6b)$$

This expression is a potential-flow closure at the exterior surface; it does not resolve the dissipative pore-mouth region.

2.4. Pore velocity and pressure averaging

Let q'_k denote the signed flow rate per unit span through channel k, positive from pore 1 to pore 2. For pore m, the surface length is

$$\ell_{m,k} = \sum_{i\in\mathcal{I}_{m,k}} S_i, \quad (7)$$

where $\mathcal{I}_{m,k}$ contains the panels within that pore. A uniform normal velocity is imposed:

$$v_{n,i} = \sigma_{m,k}\frac{q'_k}{\ell_{m,k}}, \qquad i \in \mathcal{I}_{m,k}, \quad (8)$$

with $\sigma_{1,k} = -1$, $\sigma_{2,k} = +1$ and zero normal velocity on solid panels. This mapping conserves flow through each channel; the inlet and outlet roles reverse when q'_k changes sign.

The pore-normal velocity is based on the surface pore length, whereas the mean channel velocity is based on the channel height. These velocities therefore need not be equal. Panel pressures are calculated from the local velocities before averaging over each pore:

$$p_{m,k} = \frac{1}{\ell_{m,k}}\sum_{i\in\mathcal{I}_{m,k}} p_i S_i. \quad (9)$$

Averaging is restricted to pore panels. Because the prescribed normal velocity is uniform within each pore, its squared value can be used directly in the channel energy relation.

2.5. Internal rectangular-channel model

Each channel is represented by flow between parallel plates separated by height H_k over length L_k. The mean channel velocity, hydraulic diameter and channel Reynolds number are

$$U_k = \frac{q'_k}{H_k}, \quad (10a)$$

$$D_{h,k} = 2H_k, \quad (10b)$$

$$Re_{h,k} = \frac{\rho_\infty |U_k| D_{h,k}}{\mu_\infty}. \quad (10c)$$

Starting from Bernoulli's equation with pressure losses and neglecting elevation differences, the following reduced-order channel relation is adopted:

$$p_{\mathrm{in},k} - p_{\mathrm{out},k} + \frac{1}{2}\rho_\infty v_{n,\mathrm{in},k}^2 = \Delta p_{f,k} + \frac{1}{2}\rho_\infty U_k^2. \quad (11)$$

Here, $p_{\mathrm{in},k}$ and $p_{\mathrm{out},k}$ are the averaged pressures at the higher- and lower-pressure pores, respectively, and $v_{n,\mathrm{in},k} = |q'_k|/\ell_{\mathrm{in},k}$ is the inlet pore-normal speed. The kinetic energy associated with external flow parallel to the surface is assumed to be dissipated during entry. The inlet normal contribution is retained, while the outlet contribution is approximated using the mean channel velocity with a kinetic-energy coefficient of unity.

For channels classified as laminar, the frictional pressure loss follows the parallel-plate Hagen–Poiseuille relation [20]:

$$\Delta p_{f,k} = \frac{12\mu_\infty L_k}{H_k^3}|q'_k|. \quad (12)$$

For turbulent channels, the pressure loss is calculated using the Darcy–Weisbach relation with the smooth-wall Blasius friction factor:

$$\Delta p_{f,k} = f_k \frac{L_k}{D_{h,k}}\frac{\rho_\infty U_k^2}{2}, \qquad f_k = 0.3164 Re_{h,k}^{-1/4}. \quad (13)$$

The Darcy friction-factor convention is used [21]. Extending this turbulent correlation to a parallel-plate channel through its hydraulic diameter is an engineering approximation.

The laminar relation is applied for $Re_{h,k} \le Re_{h,\mathrm{sw}}$ and the turbulent relation for $Re_{h,k} > Re_{h,\mathrm{sw}}$, where $Re_{h,\mathrm{sw}} = 2500$ is the prescribed switching Reynolds number. No blending is applied, and physical laminar–turbulent transition is not explicitly modelled. The frictional loss is zero at zero flow.

For each external-flow update, the channel relation is solved for the flow magnitude with the pore pressures and inlet normal-speed term held fixed. The direction follows the static pore-pressure difference; equal pressures give zero flow. A direct switch between the friction relations can introduce a discontinuity, so a valid channel solution must satisfy both the energy relation and the Reynolds-number condition of the selected branch.

Both friction relations assume fully developed internal flow. Entrance development and detailed flow at the pores are not resolved. Channel Reynolds numbers and length ratios $L_k/D_{h,k}$ are used to assess the scope of these approximations, as discussed in Section 2.8.

2.6. Nonlinearity and solution procedure

The panel equations are linear for prescribed pore-normal velocities. The coupled problem is nonlinear because

channel flow changes the external pressures that drive it. Nonlinearity arises from the squared velocities in the external pressure and channel energy relations, together with the nonlinear turbulent friction law. The laminar Hagen–Poiseuille relation itself is linear.

All channel flow rates are solved simultaneously by seeking agreement between the current flow estimates and the flows predicted by the channel model. The scaled residual is

$$R_k = \frac{q'_{k,\mathrm{est}} - q'_{k,\mathrm{hyd}}}{H_k V_\infty}, \tag{14}$$

where $q'_{k,\mathrm{hyd}}$ is obtained from the channel model using the current external solution. A Powell hybrid root solver seeks $\boldsymbol{R} = \boldsymbol{0}$, with a trust-region reflective least-squares method used if required [22, 23]. Each residual evaluation applies the estimated flow rates at the pores, solves the panel system and recalculates the pressure-driven channel flows. Convergence requires the scaled mismatch to satisfy the prescribed acceptance criterion.

Initial flow rates are estimated using solid-airfoil pressures and zero inlet normal velocity. Nearby converged solutions initialise subsequent cases where available. All cases reported in this study satisfy the prescribed convergence criterion. Detailed numerical settings are provided with the accompanying code.

2.7. Aerodynamic forces and pitching moment

The modelled net force includes exterior pressure forces and the reaction associated with internal channel flow. Let $\mathcal{C}$ denote the closed nominal airfoil contour, including the pore cross-sections, and $\mathcal{P}$ the union of those cross-sections. The normal points out of the enclosed airfoil volume. For steady flow, neglecting body forces, exterior skin friction and viscous tractions across the pores, the force per unit span is

$$\boldsymbol{F}' = -\int_{\mathcal{C}} (p - p_\infty)\boldsymbol{n}\,\mathrm{d}s - \rho_\infty \int_{\mathcal{P}} \boldsymbol{V} v_n\,\mathrm{d}s. \tag{15}$$

The second term accounts for momentum transport through the pores. Together, the terms include the resultant internal-wall reaction; internal-wall forces must therefore not be added again. Pressure across a pore represents a control-surface traction rather than a force on a solid surface.

Pressure and momentum flux are evaluated at the same exterior interface using $\boldsymbol{V}_i = V_{t,i}\boldsymbol{t}_i + v_{n,i}\boldsymbol{n}_i$. The dimensionless contribution from panel i is

$$\mathrm{d}\boldsymbol{C}_i = -C_{p,i}\boldsymbol{n}_i \frac{S_i}{c} - \frac{2\chi_i \boldsymbol{V}_i v_{n,i} S_i}{V_\infty^2 c}, \tag{16}$$

where $\chi_i = 1$ on pore panels and zero on solid panels. With $C_x = \sum_i \mathrm{d}C_{x,i}$ and $C_y = \sum_i \mathrm{d}C_{y,i}$,

$$\begin{aligned} C_L &= C_y \cos\alpha - C_x \sin\alpha, \\ C_{D,\mathrm{model}} &= C_x \cos\alpha + C_y \sin\alpha. \end{aligned} \tag{17}$$

The x-axis extends from the leading edge to the trailing edge, and y is positive towards the upper surface. The quarter-chord pitching moment coefficient, positive nose-up, is

$$C_M = -\sum_{i=1}^{N} \frac{(x_i - 0.25c)\,\mathrm{d}C_{y,i} - y_i\,\mathrm{d}C_{x,i}}{c}. \tag{18}$$

Pressure-only coefficients $C_{L,p}$, $C_{D,p}$ and $C_{M,p}$ can be obtained by omitting the momentum-flux term and are distinguished from the modelled net-force estimates. Exterior skin friction, separation, jet mixing and the loss-generated wake remain unresolved. Consequently, $C_{D,\mathrm{model}}$ is not a prediction of total physical drag, and lift-to-drag improvements are not inferred from this inviscid calculation.

2.8. Operating conditions and model applicability

Calculations are performed for a chord length of $c = 1$ m at $Re_c = \rho_\infty V_\infty c/\mu_\infty = 5\times10^5$. Channel heights of $H = 8$, 9 and 10 mm are examined over angles of attack from $-5°$ to $15°$. Each porous configuration is compared with the solid airfoil under the same operating conditions. XFOIL provides an additional reference for the solid-airfoil baseline [17]; it does not validate the internal channel model.

The internal-flow conditions are characterised by $Re_{h,k}$ and $L_k/D_{h,k}$, rather than by the chord Reynolds number alone. Table 1 provides space for the computed ranges across the reported cases. These quantities indicate whether short channels or proximity to the prescribed regime boundary require further examination; they do not alone establish fully developed flow.

Table 1
Channel-flow diagnostics across the reported cases. Bracketed entries must be completed from the computed results.

Configuration	Re_h range	L/D_h range
Horizontal	[insert range]	[insert range]
Vertical	[insert range]	[insert range]
Combined	[insert range]	[insert range]

The source code, channel geometry and case settings accompany the study. Numerical convergence of the coupled equations is distinguished from discretisation accuracy and physical validation.

2.9. Numerical conditions and verification

The conditions used for the reported study are listed in Table 2. The freestream speed is calculated from

$$V_\infty = \frac{Re_c \mu_\infty}{\rho_\infty c}. \tag{19}$$

The impermeable airfoil is solved with the same geometry, panel distribution and operating conditions as the channelled configurations. An inviscid XFOIL solution, based on a linear-vorticity panel method, is used as an independent reference for the impermeable-airfoil surface-pressure distribution and angle-of-attack trend [17]. This comparison is a verification of the external-flow implementation and does not validate the internal channel or pore-mouth closures.

Table 2
Numerical and operating conditions used in the present study.

Parameter	Symbol	Value
Airfoil	–	NACA 0018
Chord length	c	1 m
Baseline panel count	N_0	1000
Fluid density	ρ_∞	1.225 kg m^{-3}
Dynamic viscosity	μ_∞	1.8×10^{-5} Pa s
Freestream pressure	p_∞	101325 Pa
Chord Reynolds number	Re_c	5.0×10^5
Principal angle of attack	α	4°
Angle-of-attack range	α	−5° to 15°
Angle increment	$\Delta\alpha$	1°
Channel height	H	8, 9, 10 mm
Freestream speed	V_∞	7.3469 m s^{-1}

The baseline panel count precedes insertion of pore boundaries and any local refinement; the resulting count is recorded in each case output. Channel Reynolds numbers and length ratios identify the range over which the internal-flow approximation is applied. Detailed geometry definitions, solver settings and case-generation scripts will be made available in the accompanying GitHub repository. The archived code version and input files should correspond to the published results.

3. Results and discussion

The aerodynamic behaviour of the three porous-airfoil configurations was investigated over an angle-of-attack range of −5° to 15°. Three pore heights, $H = 8$, 9 and 10 mm, were considered for the horizontal, vertical and horizontal–vertical porous airfoils. The corresponding impermeable (solid) airfoil panel-method solution and the inviscid XFOIL prediction are included as reference cases.

The effect of the pores is first examined through the lift and moment coefficients over the angle-of-attack range. The observed aerodynamic trends are then interpreted using the corresponding pressure and velocity field differences. A detailed flow-field comparison is presented at $\alpha = 10°$ for a pore height of $H = 9$ mm.

3.1. Panel method validation against XFOIL

The impermeable source–vortex panel-method solution is compared with the corresponding inviscid XFOIL prediction in Figures. 3 and 4. Both methods show similar trends in lift and pitching moment over the investigated angle-of-attack range.

This agreement provides a useful reference for assessing the aerodynamic effects introduced by the pores. Small differences between the panel-method and XFOIL predictions are expected because the two approaches use different numerical formulations, although both represent inviscid external flow.

The impermeable panel-method solution is therefore used as the primary reference when evaluating the porous-airfoil configurations. This ensures that the porous and solid airfoils are compared using the same geometry, panel discretisation, aerodynamic formulation and operating conditions.

3.2. Effect of pores on lift

Figure 3 compares the variation of lift coefficient with angle of attack for the three porous-airfoil configurations and pore heights. The horizontal porous airfoil produces the largest change in lift. At positive angles of attack, its lift coefficient is higher than that of the impermeable reference, and the difference becomes more pronounced as the angle of attack increases. The effect also strengthens with pore height, with $H = 10$ mm producing the highest lift at larger positive angles of attack. At negative angles of attack, increasing the pore height similarly increases the magnitude of the negative lift.This nearly symmetric lift response about the zero-lift condition is consistent with the symmetric NACA 0018 geometry.

For the horizontal configuration, both ends of each pore are located on the same airfoil surface. The streamwise variation in surface pressure therefore creates a pressure difference between the two ends and drives internal flow from the locally higher-pressure region towards the lower-pressure region. This produces suction at one end and blowing at the other, changing the local surface-normal velocity and redistributing the external pressure and velocity fields along the airfoil surface. At positive angle of attack, the different pressure distributions over the suction and pressure surfaces lead to a net redistribution of aerodynamic loading that increases lift.

The stronger dependence on pore height is consistent with the reduction in hydraulic resistance as the pore becomes larger. For a given pressure difference, a larger pore height permits a greater internal flow and therefore a stronger interaction between the pore flow and the external aerodynamic field.

The vertical porous airfoil shows a distinctly different behaviour. At positive angles of attack, its lift coefficient is generally lower than that of the impermeable airfoil, while the three pore-height cases remain relatively close over most of the investigated range.

In this configuration, each pore connects the pressure surface to the suction surface. The resulting pressure difference drives internal flow between the two surfaces and partially reduces the pressure difference responsible for lift. This pressure equalisation is consistent with the lower lift coefficient obtained for the vertical porous airfoil. The small separation between the pore-height cases also indicates that this configuration is less sensitive to pore height than the horizontal arrangement.

The horizontal–vertical porous airfoil remains comparatively close to the impermeable reference. At positive angles of attack, its lift coefficient is generally slightly lower than that of the solid airfoil, and the three pore-height cases remain closely grouped.

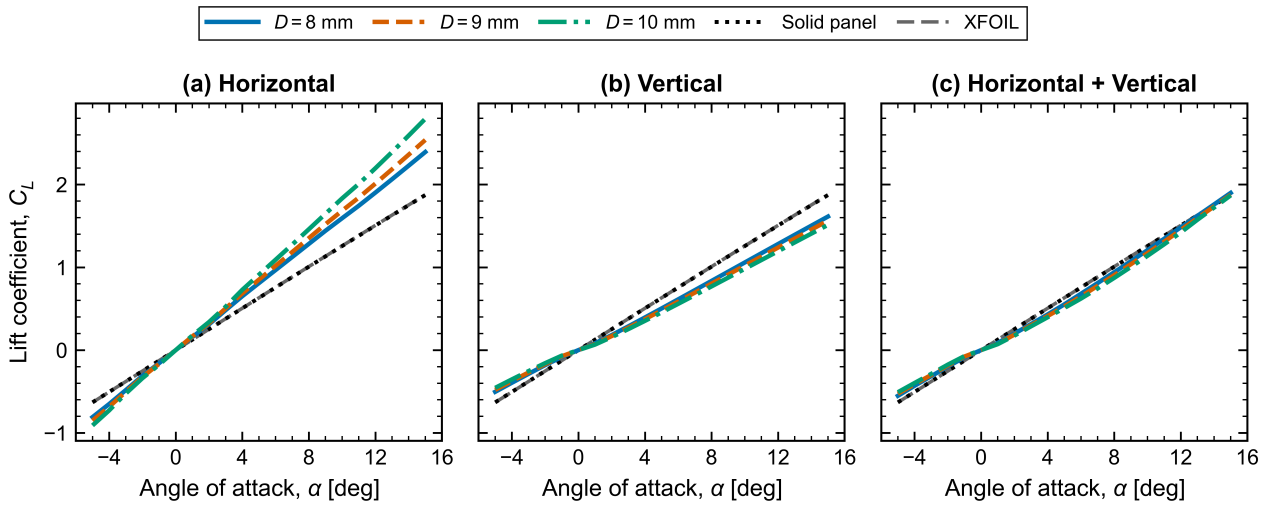


Figure 3: Lift coefficient as a function of angle of attack for (a) the horizontal porous airfoil, (b) the vertical porous airfoil, and (c) the horizontal–vertical porous airfoil. Results are shown for pore heights $H = 8$, 9 and 10 mm together with the impermeable panel-method and inviscid XFOIL reference solutions.

This behaviour reflects the simultaneous influence of the horizontal and vertical pore arrangements. The horizontal pores redistribute flow between different chordwise locations on the same surface, whereas the vertical pores transfer flow between the pressure and suction surfaces. The lift-enhancing redistribution associated with the horizontal pores is therefore partly offset by the pressure equalisation introduced by the vertical pores.

The horizontal and vertical pores remain physically separate and hydraulically independent. However, their aerodynamic effects interact through the external pressure field, since changes produced by one set of pores alter the surface pressures that drive the other set. Consequently, the response of the horizontal–vertical porous airfoil is not a simple superposition of the two individual configurations.

Overall, the results show that the aerodynamic effect of the pores is governed mainly by their orientation, placement and the pressure difference between the connected surface locations. Pore height also influences the response, but its effect is strongest for the horizontal configuration and considerably weaker for the vertical and horizontal–vertical arrangements.

3.3. Effect of pores on pitching moment

The corresponding moment coefficients are presented in Figure. 4. The horizontal porous airfoil produces the largest increase from the impermeable-airfoil response. As the positive angle of attack increases, the pitching-moment coefficient becomes progressively more negative. The magnitude of this change also increases with pore height, with $H = 10$ mm producing the largest negative pitching moment at higher positive angles of attack.

This behaviour is associated with the chordwise redistribution of surface pressure caused by the horizontal pores. Since the two ends of each pore are located at different chordwise positions along a given airfoil surface, the pressure changes induced by the internal flow act at different distances from the quarter-chord reference point. These local pressure variations therefore contribute differently to the pitching moment. Increasing the pore height strengthens the internal flow and the resulting pressure redistribution, leading to a larger change in the pitching-moment coefficient.

The increase in lift obtained with the horizontal porous airfoil is therefore accompanied by a larger negative pitching moment. This represents an important aerodynamic trade-off, since the configuration that provides the greatest lift enhancement also produces the largest change in longitudinal loading.

The vertical porous airfoil produces considerably smaller changes in moment, and the three pore-height cases remain closely grouped over most of the investigated angle-of-attack range. This behaviour is consistent with the weaker dependence on pore height observed in the lift results.

For the vertical configuration, the two ends of each pore are positioned on the upper and lower surfaces at approximately the same chordwise location. The resulting pressure redistribution therefore occurs mainly between the pressure and suction surfaces rather than between widely separated positions along the chord. The corresponding change in the chordwise pressure distribution is smaller than for the horizontal pores, resulting in a weaker pitching-moment response.

The horizontal–vertical porous airfoil exhibits behaviour between that of the individual horizontal and vertical configurations. The pitching moment becomes increasingly negative with positive angle of attack, although the departure from the impermeable-airfoil solution remains smaller than for the horizontal porous airfoil. The influence of pore height also becomes more noticeable at higher positive angles of attack.

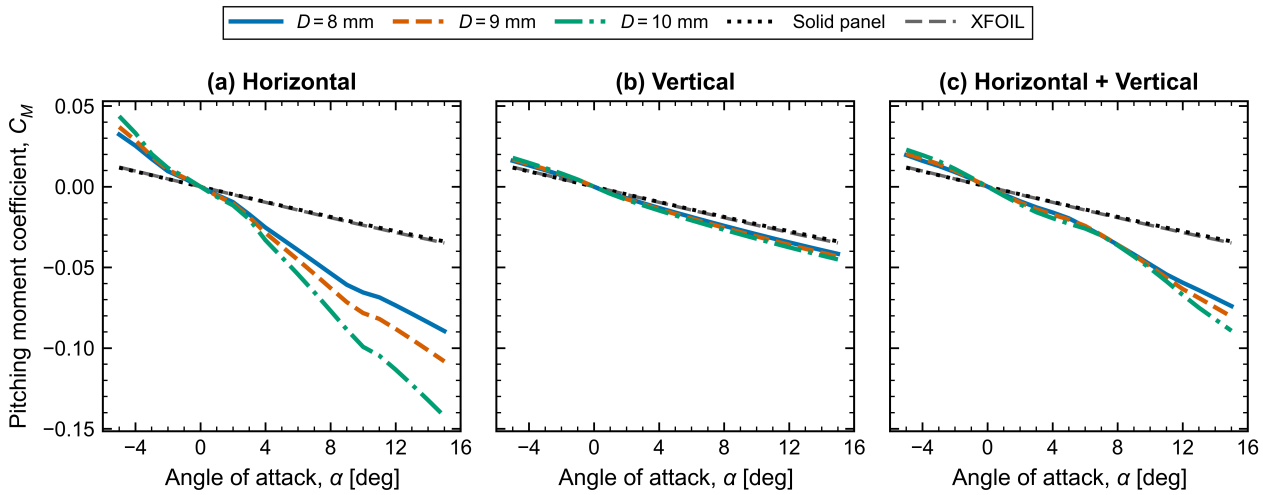


Figure 4: Pitching-moment coefficient about the quarter chord as a function of angle of attack for (a) the horizontal porous airfoil, (b) the vertical porous airfoil, and (c) the horizontal–vertical porous airfoil. Results are shown for pore heights $H = 8$, 9 and 10 mm together with the impermeable panel-method and inviscid XFOIL reference solutions.

In this configuration, the horizontal pores redistribute loading between different chordwise positions, while the vertical pores mainly redistribute pressure between the upper and lower surfaces. The two pore arrangements remain physically separate and hydraulically independent, but their aerodynamic effects interact through the external pressure field. The resulting pitching-moment variation therefore reflects the combined influence of these two pressure-redistribution mechanisms.

Overall, the pitching-moment results show that the effect of the pores depends not only on the extent of the pressure redistribution, but also on where it occurs along the chord. The horizontal porous airfoil produces the largest change in moment because the associated pressure variations occur at widely separated chordwise locations, creating a stronger moment about the quarter-chord reference point. In contrast, the vertical porous airfoil produces a smaller change because its pressure redistribution occurs mainly between the upper and lower surfaces at similar chordwise positions.

3.4. Flow field characteristics of porous airfoils

To examine how the pores alter the external flow, the pressure and velocity fields of the porous airfoils are compared with those of the corresponding impermeable airfoil at $\alpha = 10°$ and $H = 9$ mm.

The comparison is presented using pressure- and velocity-difference contours rather than the absolute flow fields. The absolute contours are dominated by the overall aerodynamic behaviour of the airfoil and therefore make the local influence of the pores more difficult to identify. Subtracting the impermeable-airfoil solution isolates the changes introduced by the pore flow, allowing the location, direction and relative strength of the resulting pressure and velocity variations to be examined more clearly.

3.4.1. Pressure field characteristics

Figure 5 presents the pressure difference between the porous and impermeable airfoils,

$$\Delta p = p_{\text{porous}} - p_{\text{solid}}, \tag{20}$$

for the three pore arrangements.

The pressure difference is shown instead of the absolute pressure field to isolate the changes introduced by the pores. In the absolute solution, the pressure distribution is dominated by the overall airfoil loading and the strong pressure gradients near the leading edge. Subtracting the impermeable-airfoil solution makes the local changes associated with the pore flow easier to identify.

The strongest pressure differences occur near the pore ends, where the internal flow directly influences the surrounding external flow. In several locations, neighbouring positive and negative values of Δp form a dipole-like pattern. This is associated with suction at one end of a pore and blowing at the other, producing local regions of reduced and increased pressure relative to the impermeable airfoil.

The effect of these pressure changes on lift depends on where they occur on the airfoil. At positive angle of attack, a reduction in pressure on the upper suction surface strengthens the suction and contributes to higher lift, whereas an increase in pressure has the opposite effect. On the lower pressure surface, an increase in pressure contributes positively to lift, while a pressure reduction tends to decrease it.

For the horizontal porous airfoil, the pressure differences are mainly concentrated around the chordwise pore ends. Because both ends of each pore lie on the same airfoil surface, the internal flow redistributes pressure between different chordwise locations. The combined pressure changes on the upper and lower surfaces increase the pressure difference across the airfoil, consistent with the higher lift coefficient observed for this configuration.

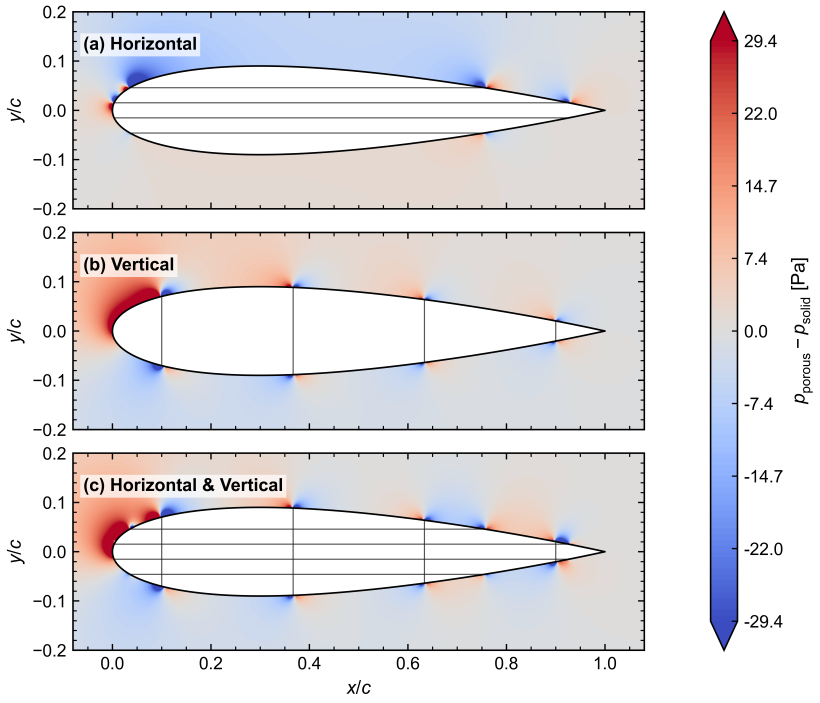


Figure 5: Pressure-field difference between the porous and impermeable airfoils at $\alpha = 10°$ and $H = 9$ mm for (a) the horizontal porous airfoil, (b) the vertical porous airfoil, and (c) the horizontal–vertical porous airfoil. Positive values indicate an increase in pressure relative to the impermeable airfoil, while negative values indicate a reduction.

The vertical porous airfoil produces a different pattern because each pore connects the pressure surface to the suction surface. Flow is therefore driven from the higher-pressure region towards the lower-pressure region, producing local pressure changes at both pore ends. At the same time, this cross-surface flow partially reduces the pressure difference between the two surfaces, which is consistent with the lower lift coefficient obtained for the vertical configuration.

The horizontal–vertical porous airfoil contains features associated with both pore arrangements. The horizontal pores redistribute pressure along the airfoil surfaces, while the vertical pores transfer flow between the pressure and suction surfaces. Although the two pore systems remain physically separate and hydraulically independent, their aerodynamic effects interact through the external pressure field. Consequently, the pressure equalisation associated with the vertical pores partly counteracts the pressure redistribution produced by the horizontal pores, giving an overall aerodynamic behaviour closer to that of the impermeable airfoil.

3.4.2. Velocity field characteristics

The velocity field is presented as the percentage difference relative to the impermeable-airfoil solution,

$$\Delta|\boldsymbol{V}|_{\%} = \frac{|\boldsymbol{V}|_{\text{porous}} - |\boldsymbol{V}|_{\text{solid}}}{|\boldsymbol{V}|_{\text{solid}}} \times 100. \tag{21}$$

This comparison isolates the changes introduced by the pores from the dominant background flow around the airfoil. The largest velocity differences occur close to the pore ends, where the imposed surface-normal flow locally perturbs the external velocity field. Adjacent regions of acceleration and deceleration are visible around several pores and correspond closely to the pressure patterns shown in Figure. 5.

The velocity and pressure fields show the expected inverse relationship described by Bernoulli's equation. Regions where the flow accelerates are generally associated with a reduction in static pressure, whereas regions of lower velocity correspond to higher static pressure. The velocity contours therefore help explain how the pore-induced changes in the local flow field lead to the observed pressure redistribution and, consequently, changes in the aerodynamic forces and pitching moment.

For the horizontal porous airfoil, alternating regions of acceleration and deceleration are concentrated around the chordwise pore ends. Since both ends of each pore lie on the same airfoil surface, the internal flow redistributes the external velocity between different chordwise locations. The combined changes on the suction and pressure surfaces strengthen the pressure difference across the airfoil, consistent with the increase in lift and the larger change in pitching moment observed for this configuration.

The vertical porous airfoil shows a different velocity pattern. Flow between the pressure and suction surfaces changes the external velocity on both sides of the airfoil and reduces part of the velocity contrast between them. This behaviour is consistent with the partial pressure equalisation discussed previously and the corresponding reduction in lift. A broader region of velocity difference is also visible near

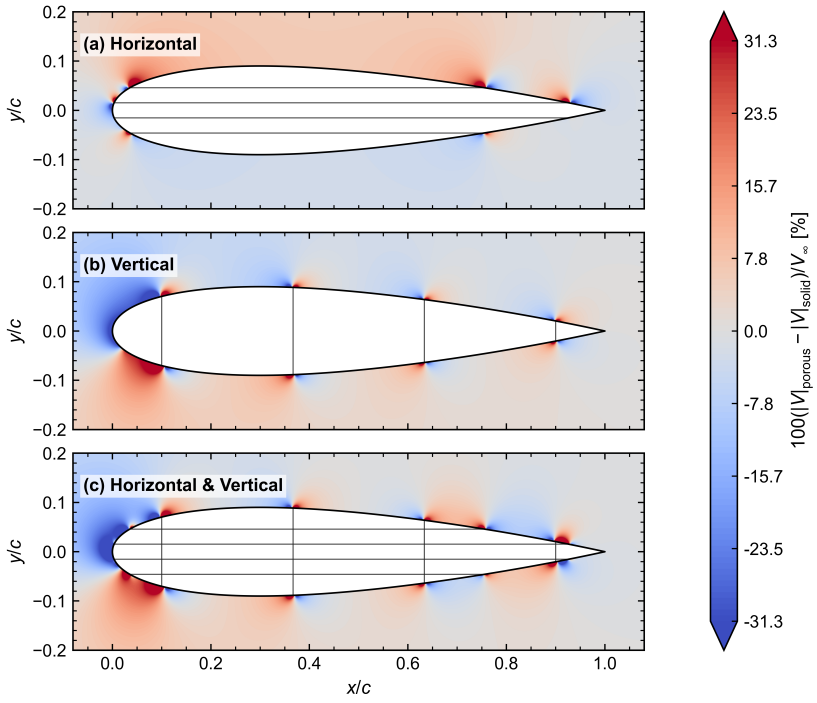


Figure 6: Percentage difference in velocity magnitude between the porous and impermeable airfoils at $\alpha = 10°$ and $H = 9$ mm for (a) the horizontal porous airfoil, (b) the vertical porous airfoil, and (c) the horizontal–vertical porous airfoil. Positive values indicate an increase in velocity magnitude, while negative values indicate a reduction relative to the impermeable airfoil.

the leading edge, where the external flow experiences strong pressure and velocity gradients.

The horizontal–vertical porous airfoil contains features associated with both pore arrangements. The horizontal pores produce local chordwise acceleration and deceleration, while the vertical pores redistribute flow between the pressure and suction surfaces. Although the two pore arrangements remain physically separate and hydraulically independent, their aerodynamic effects interact through the external flow field. The resulting velocity distribution therefore reflects the combined influence of both mechanisms rather than a simple superposition of the individual configurations.

Overall, the velocity contours support the trends observed in the pressure field and aerodynamic coefficients. The horizontal pores primarily redistribute the flow along the airfoil surfaces, whereas the vertical pores transfer flow between the pressure and suction surfaces. These different flow mechanisms contribute to the contrasting lift and pitching-moment behaviour of the three porous-airfoil configurations.

4. Conclusion and future work

A reduced-order aero–hydraulic framework has been developed to investigate NACA 0018 airfoils containing pressure-driven internal channels. The external flow is represented using a source–vortex panel method and is coupled to finite-length rectangular-channel models through the pore-normal velocity. The main conclusions of the present study are:

- The aerodynamic response of a porous airfoil depends strongly on the orientation, location and connectivity of the internal channels. Porosity alone is therefore insufficient to describe the resulting aerodynamic behaviour.
- Horizontal chordwise channels produce the strongest aerodynamic modification among the configurations considered. By connecting different chordwise locations on the same airfoil surface, these channels redistribute pressure without directly equalising the pressure and suction surfaces.
- At positive angles of attack, the horizontal configuration increases lift relative to the impermeable airfoil. The magnitude of this effect increases with channel height, demonstrating the sensitivity of the aerodynamic response to the hydraulic resistance of the internal passages.
- The lift enhancement produced by the horizontal channels is accompanied by a more negative pitching moment. This demonstrates that improvements in lift cannot be considered independently of changes in longitudinal aerodynamic loading.
- Vertical channels exhibit a fundamentally different behaviour. Connecting the pressure and suction surfaces

allows pressure-driven flow between the two sides of the airfoil, producing partial pressure equalisation and consequently reducing lift relative to the impermeable configuration at positive angles of attack.

- The combined horizontal–vertical configuration does not behave as a simple superposition of the two individual arrangements. Although the channel systems are hydraulically independent, they interact through the external pressure field. The pressure equalisation associated with the vertical channels therefore partly offsets the lift-enhancing effect of the horizontal channels.
- Pressure- and velocity-difference fields show that the strongest aerodynamic modifications occur locally around the channel openings. Suction and blowing generated by the internal flow alter the surrounding external velocity and pressure fields, which subsequently modify the integrated aerodynamic forces and pitching moment.
- The results demonstrate that the pressure difference between connected surface locations is a key driver of porous-airfoil behaviour. Channel placement, orientation and height should therefore be considered simultaneously when designing internal porous networks.
- The developed framework provides a computationally inexpensive method for screening multiple internal-channel configurations before progressing to substantially more expensive viscous CFD simulations or experimental investigations.

The present results should be interpreted as inviscid aerodynamic trends. The external panel method does not resolve boundary layers, transition, skin-friction drag, flow separation or stall, and the pressure-based streamwise-force coefficient should therefore not be interpreted as the total drag of the porous airfoil. These limitations define the next stage of the investigation.

4.1. Future work

Future development will focus on:

- incorporating a viscous boundary-layer model into the panel-method framework to estimate skin-friction drag, transition and the influence of viscous–inviscid interaction;
- extending the design space beyond the three prescribed configurations by optimising channel location, orientation, height and connectivity to identify improved internal porous-network topologies;
- investigating selected configurations using higher-fidelity unsteady CFD to resolve pore-mouth flow, boundary-layer interaction, separation and wake dynamics that cannot be represented by the present reduced-order formulation;
- validating the predicted pressure redistribution and aerodynamic trends experimentally using manufactured porous-airfoil models; and
- extending the analysis towards three-dimensional porous wings and application-relevant operating conditions once the underlying aero–hydraulic mechanisms have been established.

The present framework therefore provides the first stage of a hierarchical design process in which rapid reduced-order screening is used to identify promising internal-channel arrangements for subsequent optimisation, high-fidelity viscous simulation and experimental validation.

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